Diagnostic Inflation Expectations in China: The Role of Economic Policy Uncertainty and Geopolitical Risk

Jingle Fu

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Abstract

Household inflation expectations often respond disproportionately to salient news, generating systematic forecast errors. This paper tests whether Chinese households exhibit a *diagnostic* pattern—expectation revisions that subsequently predict forecast errors of opposite sign—using the PBoC Urban Depositors Survey. We quantify qualitative responses into a numeric expectation measure via the Carlson–Parkin method and evaluate diagnostic bias through a HAC-robust OLS baseline and a Bayesian time-varying-parameter state-space model (TVP-SSM). The OLS analysis suggests a negative relationship between forecast revisions and subsequent forecast errors, though imprecisely estimated in the small sample. The Bayesian TVP-SSM yields a negative mean diagnostic coefficient (posterior mean about -1.44 pp of forecast error per pp of revision) with wide credible intervals and time variation that we interpret descriptively. Linear drift effects of EPU/GPR are imprecise and close to zero, and we do not test nonlinearities. Finally, we use a sign-restricted Bayesian VAR to characterize a set-identified class of shocks consistent with the diagnostic overreaction-correction pattern, interpreting the results as dynamic evidence consistent with behavioral theory rather than a uniquely identified structural mechanism.

1 Introduction

The formation of inflation expectations is a cornerstone of modern macroeconomic theory and policy. While the rational expectations hypothesis remains a canonical modeling device, a wealth of empirical evidence from household and professional surveys documents persistent and systematic deviations from full-information rationality. Understanding the nature of these deviations is particularly critical for emerging economies like China, where monetary policy frameworks are evolving and household sentiment can significantly influence economic dynamics.

Chinese household inflation expectations, as recorded in the People's Bank of China (PBoC) quarterly survey, exhibit puzzling features: systematic forecast errors, volatility patterns, and episodes of sharp overreaction to transitory price shocks. These patterns invite an examination through the lens of behavioral economics. This paper investigates whether Chinese household expectations are consistent with the theory of *diagnostic expectations* (Bordalo, Gennaioli, & Shleifer, 2018). This framework, which formalizes the representativeness heuristic, posits that agents overweight salient or surprising information, leading to predictable forecast errors characterized by overreaction and subsequent correction.

China's recent economic history provides a compelling setting for this inquiry. The period from 2013 to 2025 witnessed repeated surges in economic policy uncertainty (EPU), related to domestic financial reforms and shifting growth paradigms, and spikes in geopolitical risk (GPR), most notably from U.S.-China trade tensions and global conflicts. These factors likely influence how households process information about future prices. We ask: Do Chinese households exhibit diagnostic biases in forming inflation expectations? And are these biases amplified during periods of high uncertainty or geopolitical stress?

Answering these questions requires careful empirical strategies that address two key challenges. First, one must align the quantitative measure of expected inflation derived from qualitative surveys with an appropriate measure of realized inflation, ensuring a valid test of forecast accuracy. Second, identifying a structural shock that embodies diagnostic over-

reaction requires moving beyond simple correlations to isolate disturbances that have the distinct dynamic signature predicted by the theory.

Our analysis employs two complementary Bayesian approaches to meet these challenges. First, we estimate a time-varying parameter state-space model (TVP-SSM) that directly relates forecast errors to prior forecast revisions. This model allows the key diagnostic parameter β_t to evolve over time, and we examine its association with EPU and GPR. Second, we estimate a Bayesian vector autoregression (BVAR) with sign restrictions to characterize a class of shocks consistent with the diagnostic pattern. Our identification strategy is explicitly derived from the theoretical mechanism: we search for shocks that generate an immediate rise in inflation expectations and a pattern where future inflation realizations lag behind contemporaneous expectations, thereby producing negative forecast errors. We do not impose sign restrictions on output or uncertainty; their responses are interpreted ex post.

Our findings present a coherent picture. The TVP-SSM implies a negative mean diagnostic coefficient (about -1.44 pp per pp) with wide credible intervals and time variation; the path is closer to zero early in the sample and more negative after 2018 and in the late sample. Given the short quarterly sample (N=43, $p_{\text{eff}} = 51$; Table 9), we interpret these movements as descriptive rather than causal. Linear drift coefficients on EPU/GPR are imprecise and include zero (Table 7). The BVAR analysis is consistent with this pattern by isolating a set of shocks that generate the predicted overreaction-correction dynamic. The median responses imply a large, immediate jump in expected inflation but only a muted response in realized inflation, leading to a sequence of negative forecast errors that persist for several quarters. Median responses for output and uncertainty are suggestive but imprecise. In terms of economic significance, this diagnostic news shock explains about 27% of the short-term variance in inflation expectations and around 9% of the variance in realized inflation, indicating that expectation swings are substantial while the pass-through to realized inflation remains limited in our sample.

This paper contributes to several literatures. First, it provides one of the first comprehen-

sive tests of the diagnostic expectations hypothesis in a major emerging economy, extending a framework developed primarily in advanced-economy contexts to a setting with distinct institutional and informational features. Second, methodologically, it demonstrates how sign restrictions grounded in behavioral theory can be used to characterize a set-identified class of expectation shocks consistent with the diagnostic signature, offering a template for future research. Third, it delivers new insights into the dynamics of Chinese inflation expectations, highlighting the role of behavioral biases that may complicate the transmission of monetary policy. For policymakers, our results underscore the importance of clear, consistent communication, especially during periods of high uncertainty, to mitigate salience-driven overreactions that could distort household planning and economic outcomes.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant theoretical and empirical literature. Section 3 describes our data sources, the construction of key variables—paying special attention to the alignment of forecast horizons—and presents descriptive statistics. Section 4 details the TVP-SSM and sign-restricted BVAR methodologies. Sections 5 and 6 present the core empirical results. Section 7 discusses the implications of our findings, considers alternative interpretations, and explores limitations. Section 8 concludes.

2 Literature and Theoretical Framework

2.1 Expectation Formation: From Rationality to Diagnosticity

The rational expectations hypothesis (REH), a bedrock of modern macroeconomics, posits that agents form forecasts using all available information and a (subjective) understanding of the economy's true structure, resulting in errors that are unpredictable (Muth, 1961; Lucas, 1972). This assumption imbues macroeconomic models with internal consistency and sharp policy implications. However, persistent empirical puzzles—such as the excess volatility of asset prices, the slow adjustment of professional forecasts, and systematic biases in household

surveys—have prompted economists to incorporate richer psychological foundations into models of belief formation (Shiller, 2000; Akerlof & Shiller, 2009).

Diagnostic expectations (Bordalo et al., 2018) offer a parsimonious and powerful behavioral alternative. Rooted in Kahneman and Tversky's representativeness heuristic, the theory proposes that when agents evaluate uncertain prospects, they overweight future states that are "representative" of, or similar to, recently observed signals. In a dynamic updating context, this leads to an overreaction to news. Formally, diagnostic expectations can be expressed as a distortion of standard Bayesian updating:

$$\mu_t^{\text{diag}} = \mu_t^{\text{Bayes}} + \theta \cdot (S_t - \mathbb{E}_{t-1}[S_t]),$$

where μ_t denotes the belief, S_t is a noisy signal, and $\theta > 0$ is a parameter capturing the degree of overreaction. When $\theta = 0$, beliefs are rational (Bayesian); when $\theta > 0$, agents over-extrapolate, assigning excessive weight to the surprise component of the signal.

This mechanism yields a clear empirical fingerprint: a negative correlation between forecast revisions and subsequent forecast errors. A positive revision in expectations (driven by a surprisingly high signal) should, on average, be followed by a negative forecast error as reality reverts toward its fundamental path. This prediction stands in contrast to models of informational rigidities like sticky information (Mankiw & Reis, 2002) or rational inattention (Sims, 2003), which typically generate *underreaction* and positive serial correlation in forecast errors.

Related work emphasizes information rigidities and expectation updating rules. (Coibion & Gorodnichenko, 2015) develop tests for information rigidity and document underreaction in professional forecasts, consistent with sticky or noisy information. These frameworks generate forecast errors that are positively correlated with past revisions, which is the opposite sign from the diagnostic pattern we test.

Experience-based learning offers another channel. (Malmendier & Nagel, 2016) show

that individuals overweight inflation experienced in their lifetime, yielding persistent biases without the short-horizon reversal pattern. Recent information-friction models such as (Angeletos & Lian, 2018) highlight how beliefs can respond disproportionately to news under incomplete information. Our contribution is to test the diagnostic overreaction signature in a short quarterly Chinese survey sample and to identify a class of shocks consistent with forecast-error reversals; these results complement rather than rule out information rigidity or experience effects.

2.2 Uncertainty, Geopolitical Risk, and Attention

The diagnostic expectations framework naturally interacts with the broader information environment. Economic policy uncertainty (EPU), reflecting ambiguity about future fiscal, regulatory, or monetary policy, can influence how agents process signals (Baker, Bloom, & Davis, 2016). High uncertainty may lower the signal-to-noise ratio, which in a rational model should lead to more cautious updating. However, from a behavioral perspective, heightened anxiety and media focus during uncertain times could instead amplify the salience of extreme news, potentially exacerbating diagnostic overreactions. The net effect is an empirical question.

Geopolitical risk (GPR), encompassing threats of war, terrorism, and diplomatic conflict, represents a more visceral and narrative-driven class of shocks (Caldara & Iacoviello, 2022). For an open economy like China, GPR shocks directly affect perceived inflation risks through channels such as commodity price spikes (e.g., oil, food), supply chain disruptions, and exchange rate pressure. The sudden, dramatic nature of geopolitical events may make them particularly potent triggers of the representativeness heuristic, leading households to over-infer a new regime of higher inflation from a temporary shock.

2.3 Empirical Identification and the Chinese Context

Testing for diagnostic expectations requires confronting two main tasks: establishing the reduced-form correlation pattern and, more ambitiously, identifying structural shocks that embody the theorized mechanism.

The reduced-form test is straightforward: in a regression of the forecast error $FE_t = \pi_{t+1}^{QoQ,ann} - \mu_t$ on the prior forecast revision $FR_t = \mu_t - \mu_{t-1}$, a significantly negative coefficient β is consistent with diagnostic overreaction. We acknowledge that since FE_t and FR_t both contain μ_t, measurement error in expectations could mechanically induce some negative correlation ("coupling bias"); Appendix A provides a sanity check. However, we use this regression primarily as a preliminary diagnostic signature before moving to the state-space and VAR analysis.

Structural identification is more challenging. The goal is to isolate a shock that causes expectations to move excessively relative to the eventual inflation outcome. Simply requiring that a shock raises μ_t more than $\pi_t^{QoQ,ann}$ is insufficient, as many shocks could produce this short-run pattern. The theoretical essence of a diagnostic shock is that it represents an *overreaction to fundamental news*. Therefore, our identification focuses on (1) an immediate rise in expectations and (2) a sequence of negative forecast errors, while leaving output and uncertainty responses unrestricted and interpreting them in the results. We implement this approach using sign restrictions in a BVAR.

China presents a distinctive laboratory. Households may rely more on observable price changes in frequently purchased goods (like food, particularly pork) and on state-media narratives than on central bank communications or complex macroeconomic models. The well-known "Pork Cycle" in China drives significant volatility in CPI, often dominating household perceptions of living costs regardless of core inflation trends. This context may make expectations especially susceptible to salience-driven biases, providing a strong test for the diagnostic framework.

3 Data and Variable Construction

3.1 Survey Data and Sample Coverage

Our primary data source is the Urban Depositors Survey conducted quarterly by the People's Bank of China (PBoC). This survey covers 50 major cities and samples approximately 20,000 urban households each quarter, collecting qualitative perceptions and expectations on a range of economic indicators including income, employment, housing prices, and overall prices. The key question for our purposes is: "How do you expect the price level next quarter will compare to this quarter?" Respondents choose from "rise," "basically unchanged," "fall," or "uncertain." The PBoC publishes a composite Future Price Expectations Index based on these responses, which ranges theoretically from 0 to 200, with 100 indicating neutral expectations.

Our full sample initially spans 2011Q1 to 2025Q3 for the expectations index. However, not all quarters have the detailed breakdown of responses needed for quantification. After data cleaning, we can compute quantitative expectations for 48 quarters, covering 2013Q1 to 2025Q3. The effective estimation sample for the TVP-SSM and BVAR, after accounting for the need for lagged expectations and forward inflation, runs from 2013Q4 through 2025Q2, yielding 43 quarterly observations.

It is worth noting that our sample period covers a variety of economic conditions in China: the "new normal" of slower growth around 2014-2015, the supply-side structural reforms of 2016-2018, the onset of U.S.-China trade tensions in 2018-2019, the COVID-19 shock in 2020-2021, and the post-pandemic recovery with global inflation pressures in 2022-2023. Thus, our sample encompasses both disinflationary and inflationary expectation environments, lending representativeness to our findings.

3.2 Quantifying Inflation Expectations: The Carlson-Parkin Method

The qualitative survey responses must be converted into a numeric expected inflation rate. We apply the classical (Carlson & Parkin, 1975) probabilistic method. The approach assumes each respondent has in mind a subjective probability distribution for next-period inflation, with mean μ_t and standard deviation σ_t. The respondent answers "rise" if they believe inflation will exceed some threshold U, "fall" if it will be below some threshold L, and "unchanged" if in between. Under Carlson-Parkin, one typically assumes a symmetric indifference band around zero, that is, $U = -L = \delta$, meaning if expected inflation change is within $[-\delta, +\delta]$, the respondent views it as "about the same."

Specifically, let p_t^{up} be the fraction of respondents expecting prices to increase, and p_t^{down} the fraction expecting a decrease, after excluding the "uncertain" responses. Assuming $\pi_{t+1}^{QoQ,ann} \sim N(\mu_t, \sigma_t^2)$, we have:

$$p_t^{down} = \Pr(\pi_{t+1}^{QoQ,ann} < L) = \Phi\Big(\frac{L - \mu_t}{\sigma_t}\Big), \qquad p_t^{up} = \Pr(\pi_{t+1}^{QoQ,ann} > U) = 1 - \Phi\Big(\frac{U - \mu_t}{\sigma_t}\Big).$$

With the symmetric indifference interval assumption ($U = -L = \delta$), these simplify to:

$$p_t^{down} = \Phi\Big(\frac{-\delta - \mu_t}{\sigma_t}\Big), \qquad p_t^{up} = 1 - \Phi\Big(\frac{\delta - \mu_t}{\sigma_t}\Big).$$

Define the standardized normal quantiles $z_d \equiv \Phi^{-1}(p_t^{down})$ and $z_u \equiv \Phi^{-1}(1 - p_t^{up})$, which represent the z-scores corresponding to the observed probabilities. Then the system can be written as:

$$z_d = \frac{-\delta - \mu_t}{\sigma_t}, \qquad z_u = \frac{\delta - \mu_t}{\sigma_t}.$$

Solving for μ_t and σ_t yields:

$$\sigma_t = \frac{2\delta}{z_u - z_d}, \qquad \mu_t = -\delta - \sigma_t \cdot z_d = \delta - \sigma_t \cdot z_u \ .$$

In practice, a calibration for δ is needed. The literature often either assumes a fixed indifference interval or estimates it from historical data. We follow common practice by choosing $\delta = 0.5$ percentage points, reflecting that respondents view price moves within ± 0.5 percentage points as essentially "unchanged." This choice is in line with other survey quantifications and provides a reasonable fit to historical revisions. The thresholds are symmetric around zero: respondents answering "rise" expect inflation above +0.5pp, those answering "fall" expect below -0.5pp, and "unchanged" corresponds to the interval $[-0.5, +0.5]$. Appendix Table 14 shows that varying δ in $\{0.3, 0.5, 0.7\}$ preserves the negative FE–FR correlation and negative OLS/TVP estimates, so the qualitative signature is broadly consistent; however, magnitudes attenuate as δ increases, so the calibration primarily affects scale rather than the existence of the pattern.

One additional complication is the "uncertain" responses. We handle these by first removing the "uncertain" percentage from the total and renormalizing p_t^{up}, p_t^{down}, p_t^{same} to sum to 1. Effectively, we assume respondents who answered "uncertain" have no systematic bias and their exclusion does not affect the aggregate probabilities of the others. During our sample, the "uncertain" option was chosen by about 15 percent of respondents on average, spiking above 25 percent in extreme events such as 2020Q1 at the onset of COVID-19. The high share of "uncertain" itself reflects public confusion during extreme uncertainty. In those periods, the remaining respondents might disproportionately include those with strong directional views, so our μ_t could be based on a more polarized subset. However, since the Carlson-Parkin method presumes a representative agent interpretation of the proportions, we proceed with the standard adjustment.

3.3 Aligning Expectations with Realizations: The Horizon Issue

A critical issue in testing forecast accuracy is aligning the forecast horizon. The PBoC survey asks about price changes over the **next quarter**. Therefore, to construct a commensurate measure of realized inflation, we construct the **Quarter-on-Quarter (QoQ) Annualized CPI

inflation rate** for quarter $t+1$. We derive this from the seasonally adjusted CPI index (reconstructed from monthly data). Specifically, $\pi_{t+1}^{QoQ,ann} = [(P_{t+1}/P_t)^4 - 1] \times 100$, where P_t is the price level at the end of quarter t. This measure aligns with the quarterly forecast horizon while being scaled to annual terms (percentage points) to match the standard reporting convention.

The forecast error is defined as $FE_t = \pi_{t+1}^{QoQ,ann} - \mu_t$, and the forecast revision as $FR_t = \mu_t - \mu_{t-1}$. Our final estimation sample runs from 2013Q4 to 2025Q2, yielding 43 quarterly observations.

3.4 Units and Scaling Check

It is important to clarify the units of our quantified variables. As shown in Table 1, quantified expectations μ_t have a mean of approximately 0.45 percentage points (annualized), while realized QoQ annualized inflation $\pi_t^{QoQ,ann}$ has a mean of approximately 1.75 percentage points, both measured in the same units but with different central tendencies. In the common estimation sample (Table 2), the mean forecast error is 1.13 percentage points, implying that realized $\pi_{t+1}^{QoQ,ann}$ exceeded quantified expectations μ_t on average. The fixed indifference parameter $\delta = 0.5$ in the Carlson-Parkin method acts as a threshold for "unchanged" responses.

3.5 Uncertainty, Risk, and Control Variables

We incorporate the China Economic Policy Uncertainty Index (Baker et al., 2016) and the Global Geopolitical Risk Index (Caldara & Iacoviello, 2022), averaging their monthly values to quarterly frequency. The China EPU index is constructed via news article counts in the South China Morning Post. In our sample the EPU index has several notable spikes: it jumped during the 2015 stock market crash and RMB exchange rate reform, reached a peak during the initial U.S.-China trade dispute in 2018, and spiked again in early 2020 during COVID-19. The global GPR index captures worldwide geopolitical risk events. It spikes

around events like the 2014 Crimea crisis, the 2017 North Korea missile crisis, and reaches its highest value in early 2022 with the Russia-Ukraine war.

To account for other potential determinants of forecast errors, we include control variables in some specifications: industrial value added growth (a proxy for the business cycle), broad money (M2) growth (controlling for monetary conditions and liquidity), producer price index (PPI) inflation (upstream price pressure), and food CPI inflation. Food prices—especially pork in China—often drive short-term CPI swings and consumer sentiment, representing a particularly salient channel through which households form expectations. All data are sourced from the PBoC, the National Bureau of Statistics of China, and the cited index creators.

3.6 Descriptive Evidence

Table 1 and Table 2 present summary statistics. Several features stand out. First, in the common estimation sample (Table 2), the mean forecast error is 1.13 percentage points, indicating that, on average, actual inflation turned out higher than households expected. This suggests a systematic under-prediction bias in our sample period. This finding differs from some international evidence showing over-estimation, and may reflect China's specific context: households anchoring expectations to the low-inflation "new normal" period of the mid-2010s, subsequently surprised by food price shocks (pork cycle) and global inflation spillovers post-2020. Second, the uncertainty and risk indices exhibit large variation (EPU in particular varies by a factor of six from trough to peak), mirroring the turbulent economic and geopolitical environment of the 2013-2025 period.

Most importantly, there is a negative correlation between FE_t and FR_t ($r = -0.19$, as detailed in Table 4), providing initial evidence that when expectations go up, subsequent errors tend to be negative, and vice versa. A basic OLS regression (Table 3) yields a negative coefficient on forecast revisions, which is marginally significant at the 10% level in specifications including control variables. While the relationship is imprecisely estimated given the

limited sample size (N=43), the sign is consistent with diagnostic overreaction. This regression, albeit simple, foreshadows our main results: a negative coefficient on forecast revisions is exactly what diagnostic expectations theory predicts. We turn to Bayesian methods for more precise inference.

Table 1: Descriptive Statistics of Key Variables (Quarterly)

variable	count	mean	std	min	25%	median	75%	max
infl_exp_index	58.000	62.862	4.294	56.100	59.650	61.800	64.375	74.800
μ_{cp}	48.000	0.452	0.551	-1.287	0.120	0.411	0.704	1.736
CPI_QoQ_Ann	58.000	1.746	3.266	-6.679	-0.514	1.938	4.049	8.314
Food_CPI_YoY_qavg	58.000	3.498	5.276	-5.148	-0.342	2.665	5.458	20.299
epu_qavg	58.000	225.325	112.921	75.909	125.531	222.375	306.892	499.249
gpr_qavg	58.000	105.028	29.137	69.691	87.027	95.891	118.628	224.596
Ind_Value_Added_YoY	58.000	5.867	3.152	-0.367	3.958	5.883	7.017	13.933
M2_YoY	58.000	10.893	2.536	6.480	8.542	10.950	13.033	16.510

Mean/Std calculated using available sample. See data appendix for variable definitions.

Table 2: Forecast Error and Revision Statistics

Variable	N	Mean	Std	Min	P25	Median	P75	Max
Forecast Error (FE)	43.000	1.130	3.171	-7.004	-0.670	1.453	3.110	7.611
Forecast Revision (FR)	43.000	-0.051	0.507	-1.612	-0.306	-0.042	0.319	0.916
Corr(FE, FR)	43.000	-0.192						

$FE_t = CPI_{t+1} - \mu_t$ (Forecast Error).
$FR_t = \mu_t - \mu_{t-1}$ (Forecast Revision).
$Corr(FE, FR) < 0$ suggests diagnostic expectations.

Table 4 presents the pairwise correlations among our key variables. The quantified expectation μ_t correlates positively with CPI_QoQ_Ann inflation (labeled "CPI" in the table), but this correlation is far from perfect, confirming that expectations contain substantial variation independent of actual outcomes. As shown in the table, the negative correlation between FE_t and forecast revision FR_t provides the key diagnostic signature. EPU shows a moderately negative correlation with μ_t, possibly because EPU surges often coincide with deflation fears or demand collapse scenarios. GPR's correlations are generally smaller in magnitude, suggesting geopolitical risk has more idiosyncratic effects not strongly captured by linear associations.

Table 3: Diagnostic Expectations: OLS Baseline (Forecast Errors on Revisions)

Variable	(1)	(2)	(3)
Constant	1.069***	1.296***	-6.103
	(0.371)	(0.418)	(8.479)
Forecast Revision (FR_t)	-1.200	-1.296*	-1.325*
	(0.780)	(0.774)	(0.729)
Food CPI YoY		-0.080	-0.054
		(0.078)	(0.090)
M2 Growth YoY			-0.057
			(0.190)
PPI YoY			0.079
			(0.074)
Observations	43		
Observations		43	
Observations			43
R^2	0.037	0.054	0.070
Adj. R^2	0.013	0.007	-0.028

Dependent variable: $FE_t = \pi_{t+1}^{QoQ,ann} - \mu_t$ (forecast error), where $\pi^{QoQ,ann}$ is CPI QoQ Ann.
Key regressor: $FR_t = \mu_t - \mu_{t-1}$ (forecast revision).
Newey-West HAC standard errors (lag=4).
Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
If $\beta_{FR} < 0$ and significant, it supports diagnostic expectations (overreaction).

Table 4: Correlation Matrix (Key Variables)

Variable	μ_{CP}	CPI	FE	FR	Food	EPU	GPR	M2	PPI
μ_{CP}	1.000								
CPI	0.537	1.000							
FE	-0.207	-0.483	1.000						
FR	0.453	0.649	-0.192	1.000					
Food	0.139	0.199	-0.109	-0.116	1.000				
EPU	-0.458	-0.032	0.176	0.011	0.185	1.000			
GPR	-0.169	-0.039	-0.078	0.040	-0.421	0.097	1.000		
M2	0.343	-0.040	-0.078	-0.077	0.185	-0.456	-0.156	1.000	
PPI	-0.063	0.089	0.143	0.060	-0.334	0.140	0.015	-0.345	1.000

Lower triangle shown.
Based on common sample.

Table 5 examines whether diagnostic expectations behavior differs across distinct macroeconomic regimes by splitting the sample into pre-2018 and 2018-onward subperiods, using the μ_t-available sample that starts in 2013Q1. The year 2018 marks an important structural break when U.S.-China trade tensions escalated sharply. Comparing the two subperiods, we observe striking differences. Quantified expectations μ_t were notably higher in the 2013-2017 period compared to 2018-2025, reflecting that Chinese inflation has been generally lower and more stable in recent years. However, the mean forecast error ($FE_t = \pi_{t+1}^{QoQ,ann} - \mu_t$) remains positive in both subperiods (0.90 in 2013-2017, 0.98 in 2018-2025), indicating persistent under-prediction on average. The subperiod differences are more pronounced in dispersion and tail realizations rather than in the mean, suggesting that regime changes may affect the volatility of errors and revisions more than their unconditional direction. Crucially, uncertainty measures jumped dramatically in the post-2018 period, with EPU nearly tripling and GPR also rising substantially. These increases underscore that the post-2018 period was characterized by unprecedented policy and geopolitical uncertainty. This sets the stage for understanding why a time-varying approach is necessary: the relationship between expectation updates and subsequent errors adapts to the macroeconomic environment.

Table 5: Sub-period Statistics: Pre vs Post 2018

Period	Variable	N	Mean	Std	Min	Max
2013-2017 (μ sample)	mu_cp	18.000	0.793	0.491	0.171	1.736
2013-2017 (μ sample)	CPI_QoQ_Ann	18.000	1.892	3.007	-4.322	6.825
2013-2017 (μ sample)	FE	18.000	0.896	3.044	-5.749	6.055
2013-2017 (μ sample)	FR	15.000	-0.075	0.418	-0.902	0.650
2013-2017 (μ sample)	epu_qavg	18.000	128.780	50.384	75.909	277.163
2013-2017 (μ sample)	gpr_qavg	18.000	99.150	17.208	73.702	135.585
2018-2025 (μ sample)	mu_cp	30.000	0.247	0.485	-1.287	1.270
2018-2025 (μ sample)	CPI_QoQ_Ann	30.000	1.443	3.292	-6.679	8.314
2018-2025 (μ sample)	FE	29.000	0.980	3.337	-7.004	7.611
2018-2025 (μ sample)	FR	29.000	-0.010	0.566	-1.612	0.916
2018-2025 (μ sample)	epu_qavg	30.000	305.227	71.802	165.772	453.896
2018-2025 (μ sample)	gpr_qavg	30.000	114.319	35.227	69.691	224.596

Subperiods based on quarters with non-missing μ_{cp} (2013Q1-2025Q3).
2018 marks the escalation of US-China trade tensions.
EPU and GPR means rose in the post-2018 period.

4 Empirical Methodology

4.1 A Time-Varying Parameter State-Space Model (TVP-SSM)

To model the evolving relationship between forecast errors and revisions, we specify a state-space model. The measurement equation is:

$$FE_t = \alpha + \beta_t FR_t + \gamma' Z_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, R), \tag{1}$$

where Z_t is a vector of control variables (food inflation, M2 growth, PPI inflation). The time-varying coefficient β_t captures the diagnostic bias. A negative β_t indicates that a positive forecast revision predicts a negative future error.

The state equation governs the evolution of β_t:

$$\beta_t = \beta_{t-1} + d_{\text{EPU}} \Delta\text{EPU}_t + d_{\text{GPR}} \Delta\text{GPR}_t + u_t, \qquad u_t \sim \mathcal{N}(0, Q). \tag{2}$$

This allows the bias to follow a random walk with a drift component that depends on changes in the uncertainty indices ($\Delta EPU_t, \Delta GPR_t$). The coefficients d_{EPU} and d_{GPR} test whether rising uncertainty linearly amplifies the diagnostic bias.

We estimate the model using Bayesian methods. Priors are weakly informative and provide regularization given the short sample: $\beta_0 \sim \mathcal{N}(0, 1^2)$, $d_{\text{EPU}}, d_{\text{GPR}} \sim \mathcal{N}(0, 100^2)$, and $R, Q \sim \mathcal{IG}(2, 1)$. Appendix B reports prior sensitivity. The Gibbs sampler iterates between drawing the latent state path $\{\beta_t\}$ via the forward-filtering backward-sampling algorithm and drawing the static parameters conditional on the states.

4.2 Identifying a Diagnostic News Shock with Sign Restrictions

To move beyond correlations and explore the dynamic macroeconomic effects of a diagnostic bias, we estimate a Bayesian VAR and use sign restrictions to characterize a class of shocks consistent with diagnostic overreaction (Uhlig, 2005). The VAR includes five variables:

$y_t = (\mu_t, \pi_t^{QoQ,ann}, g_t, EPU_t, GPR_t)'$, where g_t is industrial production growth. We include two lags and use a diffuse conjugate Normal-Inverse-Wishart prior ($B_0 = 0$, $V_0 = 10I$, $S_0 = I$, $\nu_0 = n + 2$).

Let the reduced-form VAR be $y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \epsilon_t$, with $\epsilon_t \sim \mathcal{N}(0, \Sigma)$. A structural shock is identified via an orthogonalization $\epsilon_t = B\eta_t$, where $\eta_t \sim \mathcal{N}(0, I)$ and B satisfies $BB' = \Sigma$. Any $B = PQ$ is valid, where P is the Cholesky factor of Σ and Q is an orthonormal matrix.

We search for a column of B (a shock) whose impulse responses satisfy a set of sign restrictions designed to capture a "diagnostic news shock." The key theoretical insight is that we seek to identify a disturbance that generates overreaction in expectations to news about future economic conditions, rather than simply any shock that raises expectations. Simply requiring that a shock raises μ_t more than $\pi_t^{QoQ,ann}$ would be insufficient, as many shocks could produce this short-run pattern. The theoretical essence of a diagnostic shock is that it represents an overreaction to fundamental news.

Therefore, our identification strategy imposes restrictions that distinguish a diagnostic shock from alternative interpretations such as a pure noise shock or a conventional demand shock. Specifically, our baseline restrictions are:

- **On impact ($h = 0$):** $IRF_\mu(0) > 0$.
- **For horizons $h = 0, 1, 2, 3$:** $IRF_{\pi^{QoQ,ann}}(h+1) - IRF_\mu(h) < 0$.

The first set requires the shock to initially raise inflation expectations. The second set imposes the diagnostic signature: expected inflation in any period exceeds the realized inflation in the next period, creating negative forecast errors. We leave the responses of g_t, EPU_t, and GPR_t unrestricted, allowing the data to reveal whether such shocks occur with real expansion or uncertainty resolution. Because we do not impose $IRF_{\pi^{QoQ,ann}}(0) > 0$ or $IRF_g(0) > 0$, the impact bands for CPI and output in Table 10 can cross zero. For each posterior draw, we search over random rotations and retain the first shock column that

satisfies these restrictions (with sign normalization), so the IRF summaries are computed across accepted draws only.

These restrictions define a set-identified class of shocks consistent with the diagnostic forecast-error signature. Imposing the forecast-error pattern for multiple horizons, not just on impact, anchors identification in the dynamic implication of the theory rather than a single contemporaneous condition. The extremely high acceptance rate (over 99.95 percent) indicates that the restrictions are relatively permissive and does not establish that the identified shocks are uniquely structural or fully representative. We therefore interpret the resulting IRFs as dynamic patterns within this identified set, not as causal proof of a single underlying mechanism.

Table 6: BVAR sign-restriction acceptance

accepted draws	posterior draws	rotations	total rotations	accept rate per draw
1999	2000	200	5152	0.999500

A posterior draw is accepted if at least one rotated structural shock (any column, with sign normalization) satisfies the DE-shock restrictions.

This strategy aims to characterize shocks that are consistent with an expansionary interpretation (when output responds positively) but to which households overreact in their inflation forecasts. We draw 2,000 posterior draws of the VAR parameters. For each draw, we randomly sample orthonormal matrices Q up to a cap of 200 rotations (default `subdraws=200`) and stop early once a rotated column satisfies the restrictions. Table 6 reports the actual total number of rotations attempted across draws (which can be well below the cap because of early stopping). The median impulse responses are reported in standardized units (the VAR is estimated using standardized variables).

5 Results: The Time-Varying Diagnostic Bias

5.1 The Path of Diagnostic Intensity

Figure 1 plots the posterior mean of β_t over time along with its 90% credible interval. Recall that β_t is the coefficient on the forecast revision FR_t in the measurement equation: a negative β_t indicates that when households raised their expected inflation ($FR_t > 0$), the subsequent forecast error FE_t tended to be negative, which is the signature of diagnostic expectations.

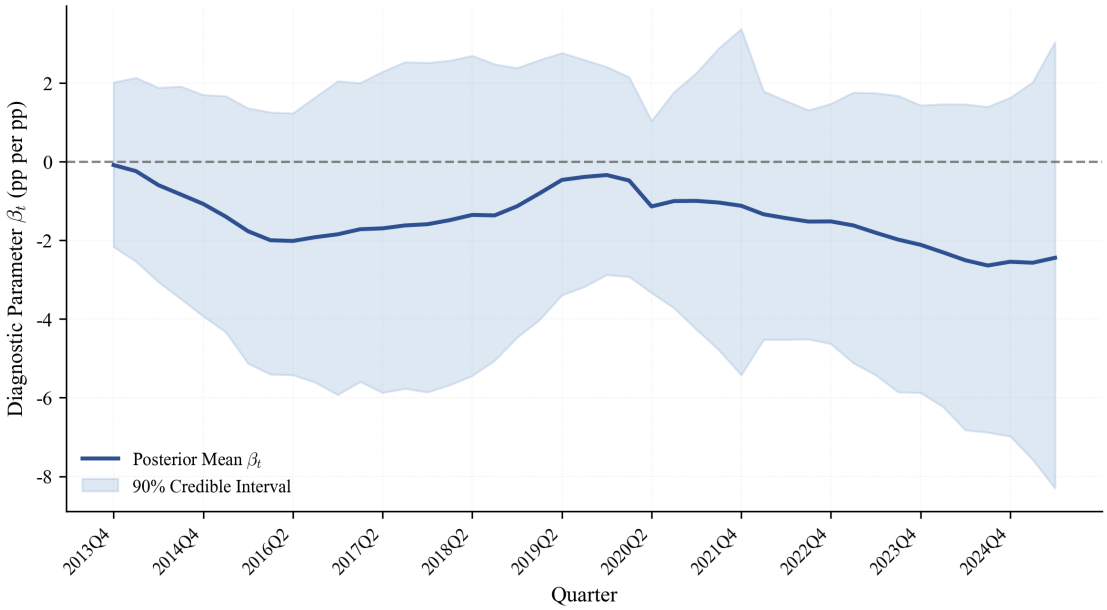

Figure 1: Estimated Time Path of Diagnostic Expectations Parameter β_t (2013Q4–2025Q2)
Note: The solid line is the posterior mean of β_t, and the shaded region is the 90% credible interval. FR_t and FE_t are in percentage points (QoQ annualized) and are not standardized, so β_t is interpreted as pp of forecast error per pp of forecast revision. $\beta_t < 0$ indicates diagnostic overreaction. Credible bands are wide, so the time variation is interpreted descriptively.

We observe several features in the β_t path. The posterior mean is negative in every quarter, with an average of about -1.44. The 90 percent credible bands are wide and often include zero, so we interpret the magnitude cautiously, but the mean path indicates persistent overreaction.

Time variation is nonetheless evident. The path is closer to zero in the early 2013–2017 period and becomes more negative after 2018, with the late-sample quarters showing

the most negative means. This comparison highlights broad regime differences rather than event-by-event matches.

Given the short quarterly sample (N=43) and the large effective parameter count of the TVP specification ($p_{\text{eff}} = 51$ in Table 9), we treat the β_t path as descriptive and avoid causal interpretations of individual swings.

5.2 Drivers of Time Variation: EPU and GPR

The state equation allows us to test whether observed uncertainty measures linearly drive changes in β_t. The posterior estimates for the drift coefficients are $d_{\text{EPU}} = 0.071$ (90% credible interval: $[-0.354, 0.482]$) and $d_{\text{GPR}} = -0.023$ (90% interval: $[-0.665, 0.600]$). While the point estimate for d_{GPR} is negative—suggesting that rising geopolitical risk might intensify the bias—and the point estimate for d_{EPU} is positive, the credible intervals for both are wide and include zero. This indicates that, in our sample, a simple linear relationship between quarterly changes in these indices and shifts in diagnostic intensity is not precisely estimated.

This null result indicates that, in our sample, a simple linear relationship between quarterly changes in these indices and shifts in diagnostic intensity is not precisely estimated. The scatter plots in Figure 2 and Figure 3 are therefore descriptive summaries rather than evidence of a systematic linear link. Given the short time series, we do not implement formal nonlinear or threshold tests; we leave such tests for future work.

5.3 Measurement Equation and Model Fit

Table 7 reports posterior summaries for the static parameters and drift terms (90% credible intervals; standardized controls). The intercept α is positive, while the point estimates on food inflation and M2 growth are negative but imprecise, with 90% intervals that include zero. The PPI coefficient is likewise imprecise. The drift terms d_{EPU} and d_{GPR} are centered near zero with wide intervals, consistent with the weak linear association emphasized above.

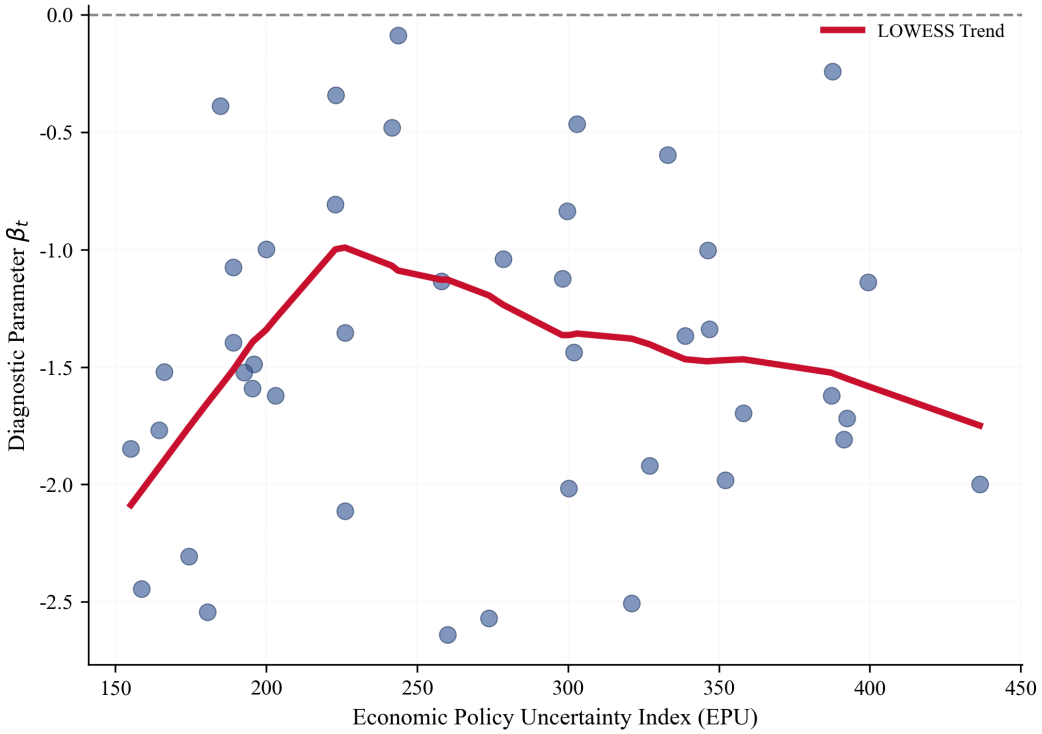

Figure 2: Scatter Plot: β_t versus Economic Policy Uncertainty

Note: Each point represents one quarter. The fitted line is flat and imprecise, and the scatter is wide. We treat this plot as descriptive and do not test nonlinear or threshold effects.

Table 7: Bayesian State-Space Model: Posterior Summary of Parameters

param	mean	sd	p05	p95
alpha	1.072	0.477	0.287	1.843
gamma_Food_CPI_YoY_qavg	-0.261	0.516	-1.103	0.588
gamma_M2_YoY	-0.100	0.507	-0.933	0.716
gamma_PPI_YoY	0.362	0.526	-0.491	1.231
d_epu_qavg	0.071	0.254	-0.354	0.482
d_gpr_qavg	-0.023	0.383	-0.665	0.600
R_var	9.721	2.280	6.626	13.887
Q_var	1.003	0.861	0.276	2.500

Z are standardized control variables (includes food inflation control); X are standardized uncertainty proxies (EPU/GPR).
R_var is measurement error variance, Q_var is state innovation variance.

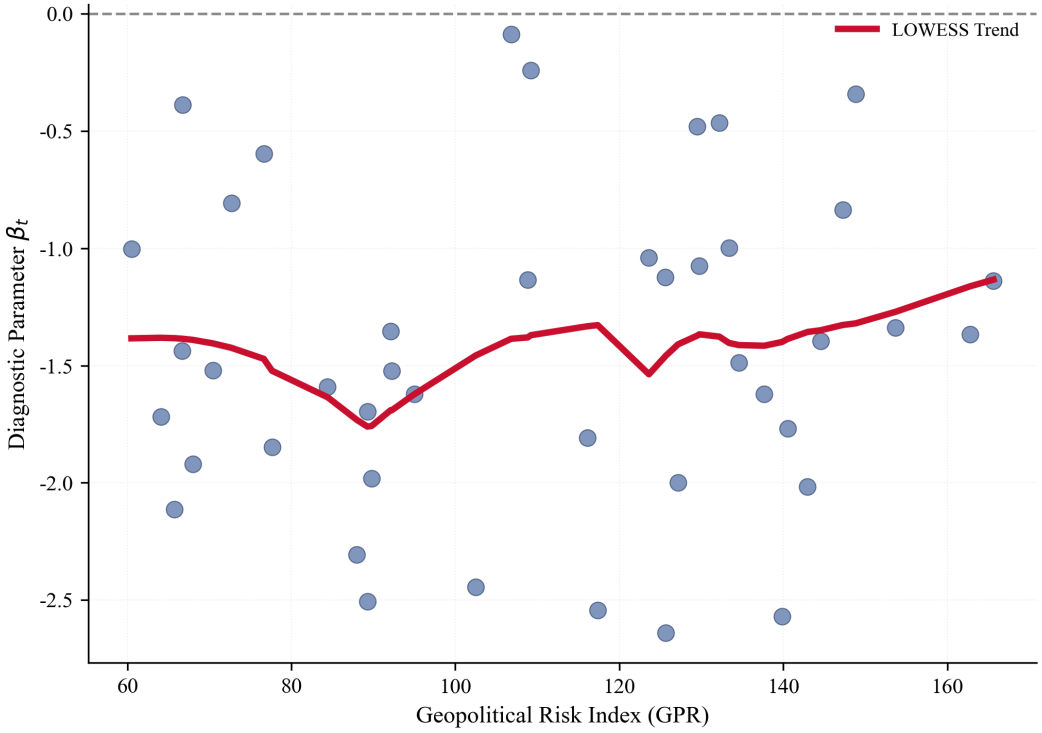

Figure 3: Scatter Plot: β_t versus Geopolitical Risk

Note: Higher GPR levels are not tightly related to β_t in a linear specification; the scatter is wide and the fitted slope is imprecise. We treat this plot as descriptive.

5.4 MCMC Convergence Diagnostics

To ensure the reliability of our Bayesian estimates, we conduct standard MCMC convergence diagnostics. We run four chains of 10,000 iterations each, discarding the first 3,000 as burn-in. Table 8 reports Geweke z-statistics (comparing means in the first and last portions of the chain), Gelman-Rubin potential scale reduction factors $\hat{R}$, and effective sample sizes ($ESS > 300$ in all cases). All Geweke statistics are within ± 1.96, indicating no significant drift in the chains. The $\hat{R}$ values are all at or very close to 1.00, confirming between-chain convergence. Effective sample sizes range from about 325 to 7,000.

Visual inspection of trace plots and autocorrelation functions further confirms convergence: trace plots exhibit stable, well-mixed chains without trends, and ACF plots show rapid decay. The posterior distributions of all parameters are well-defined and unimodal. These diagnostics collectively validate that our posterior inference is not an artifact of poor sampling but reflects the true posterior distribution conditional on the model and data.

Table 8: MCMC Convergence Diagnostics

Parameter	Geweke_z	Geweke_p	R_hat	ESS	Converged
alpha	-1.171	0.242	1.0000	7000	yes
gamma[0]	0.763	0.446	1.0000	7000	yes
d[0]	-1.953	0.051	1.0003	618	yes
R	-0.662	0.508	1.0001	5226	yes
Q	0.993	0.321	1.0002	325	yes

Geweke: H0 is chain converged (first 10% vs last 50% equal means); $|z| < 1.96$ passes.
Gelman-Rubin: Based on 4 independent chains, $\hat{R} < 1.1$ (conservative < 1.05) passes.
ESS: Effective Sample Size; $ESS > 300$ is the threshold (relaxed for variance parameters in small samples).
Conclusion: Converged if Geweke, R-hat, and $ESS > 300$ all satisfy criteria.

Additionally, we compare models using total-sample log-likelihood evaluated at point estimates and the AIC/BIC (Table 9). The TVP-SSM delivers the highest in-sample log-likelihood, but the information criteria penalize its large parameter count (p_eff includes the time-varying β_t path). We interpret this trade-off as follows: while the TVP specification

does not claim superior out-of-sample predictive performance, it serves as a descriptive tool to recover the latent path of β_t, revealing significant temporal heterogeneity that constant-parameter models mask. The information criteria caution against over-parameterization, but the improved in-sample fit supports the time-varying approach for our research question.

Table 9: Model Comparison: Log-Likelihood and Information Criteria

Model	log_lik	AIC	BIC	p_eff	Notes
OLS (FR only)	-109.3	224.7	229.9	3.0	Gaussian OLS baseline
SSM (fixed β)	-108.6	229.2	239.7	6.0	OLS with controls
TVP-SSM (EPU+GPR)	-107.1	316.1	406.0	51.0	Posterior-mean TVP

log_lik is total-sample log-likelihood evaluated at point estimates.
For TVP-SSM, log_lik uses posterior-mean parameters and the β_t mean path.
Controls are standardized as in the TVP-SSM; FE/FR are in pp units.
AIC/BIC computed from log_lik and p_eff; p_eff counts time-varying β_t for TVP.
TVP-SSM parameters use posterior means from Table 7 (R_var=9.721, Q_var=1.003).

6 Results: The Dynamics of a Diagnostic News Shock

6.1 Identification Success and Shock Properties

Our sign restriction algorithm achieves a high acceptance rate of 99.95 percent (Table 6). This high rate indicates that the imposed restrictions define a permissive identified set rather than sharply pinning down a unique structural mechanism. The resulting impulse responses should be interpreted as characterizing the dynamics of a broad class of shocks consistent with the diagnostic pattern (expectations rising more than subsequent inflation), rather than a single causal shock. We report the median responses across this identified set along with credible bands that reflect both parameter uncertainty and set identification.

6.2 Impulse Response Analysis

Figure 4 presents the median impulse responses to a one-standard-deviation diagnostic news shock, along with 68 percent credible bands. All responses are in standardized units. Table

10 reports quantitative IRF values at selected horizons.

Table 10: Impulse responses to the identified DE shock (standardized units)

variable	h	p16	p50	p84
μ_{cp}	0	0.192500	0.349200	0.506200
μ_{cp}	1	0.075900	0.194800	0.333900
μ_{cp}	2	0.061900	0.140300	0.233400
μ_{cp}	4	0.014800	0.095300	0.186100
μ_{cp}	8	-0.004800	0.038400	0.109000
μ_{cp}	12	-0.006600	0.016900	0.070700
CPI_QoQ_Ann	0	-0.143500	0.087800	0.331200
CPI_QoQ_Ann	1	-0.107900	0.041600	0.181400
CPI_QoQ_Ann	2	-0.140600	-0.029800	0.081800
CPI_QoQ_Ann	4	-0.063400	0.002400	0.062100
CPI_QoQ_Ann	8	-0.036700	-0.003800	0.023200
CPI_QoQ_Ann	12	-0.022700	-0.002400	0.010600
Ind_Value_Added_YoY	0	-0.112800	0.138400	0.374200
Ind_Value_Added_YoY	1	-0.077600	0.062200	0.204800
Ind_Value_Added_YoY	2	-0.053200	0.048500	0.160100
Ind_Value_Added_YoY	4	0.005700	0.078900	0.166100
Ind_Value_Added_YoY	8	-0.004600	0.031800	0.090500
Ind_Value_Added_YoY	12	-0.005200	0.014900	0.057000
epu_qavg	0	-0.398700	-0.164800	0.104900
epu_qavg	1	-0.316700	-0.166600	0.009300
epu_qavg	2	-0.274400	-0.131700	0.021800
epu_qavg	4	-0.212800	-0.098800	0.012600
epu_qavg	8	-0.149000	-0.050300	0.012200
epu_qavg	12	-0.106200	-0.024300	0.010200
gpr_qavg	0	-0.391400	-0.023300	0.364700
gpr_qavg	1	-0.176000	0.048700	0.273000
gpr_qavg	2	-0.193300	-0.051600	0.094700
gpr_qavg	4	-0.120500	-0.027500	0.063600
gpr_qavg	8	-0.053400	-0.007700	0.033300
gpr_qavg	12	-0.027400	-0.002700	0.019500

Median and 16/84

Expectations and Inflation. Expected inflation (μ_t) shows a substantial increase on impact, with the response persisting for about two years before fading. In stark contrast, the response of realized inflation ($\pi_t^{QoQ,ann}$) is muted and uncertain, with a wide credible band that includes zero. The CPI_QoQ_Ann response subsequently turns slightly negative and

remains negligible thereafter. By construction under the sign restrictions, the forecast-error IRF $FE_h = IRF_{\pi^{QoQ,ann}}(h+1) - IRF_{\mu}(h)$ is negative for the first several quarters. Figure 5 visualizes the forecast error path, which remains substantially negative through multiple quarters before partially reversing.

Macroeconomic Context. The median responses suggest an expansionary profile. Industrial production growth (g_t) responds positively in the median, with an impact response of 0.16 standard deviations (band: [-0.09, 0.40]), though the credible band is wide and includes zero at short horizons, so the expansionary interpretation should be viewed as suggestive rather than definitive. Both the EPU and GPR indices *decline* following the shock in the median. EPU falls by a median of 0.17 standard deviations on impact (band: [-0.40, 0.08]), but the response is imprecisely estimated; GPR similarly drops, though with greater uncertainty (median: -0.03, band: [-0.41, 0.35]). These median declines are consistent with episodes of uncertainty resolution or positive surprises—moments when favorable news arrives and households overextrapolate the good developments. This interpretation aligns with the theoretical notion that diagnostic expectations overweight salient recent signals: when good news arrives and uncertainty partially resolves, households may become overly optimistic about sustained inflation. However, given the wide credible bands, we interpret the co-movement with uncertainty as indicative rather than conclusive.

6.3 Variance Decomposition and Economic Significance

Table 11 reports the forecast error variance decomposition (FEVD) for key variables at selected horizons. The diagnostic news shock is a major driver of fluctuations in inflation expectations, accounting for about 26–28 percent of their forecast error variance across horizons. This is a substantial share, highlighting the quantitative importance of this overreaction channel for understanding expectation volatility.

Crucially, the shock explains a much smaller fraction—about 9 percent—of the variance in realized inflation. It also contributes modestly (roughly 11–12 percent) to output fluctuations

and explains a notable portion of the movements in the uncertainty indices, with EPU's share increasing from about 12 percent at the 1-quarter horizon to about 20 percent at the 3-year horizon. This decomposition reveals an important asymmetry: while diagnostic shocks cause large swings in what people *expect* to happen to prices, they have a limited direct role in driving what actually *happens* to prices. This asymmetry shows that our sign restrictions do not force the identified shock to explain all macroeconomic variation, but instead isolate a *specific* behavioral shock. The fact that it is a primary driver of household expectation volatility but not realized inflation suggests that, in our sample, the overreaction in expectations did not become self-fulfilling to a significant degree, perhaps due to anchoring from other agents (firms, markets) or offsetting policy actions.

Table 11: Forecast Error Variance Decomposition: Diagnostic Expectations Shock

Variable	Horizon	FEVD_%
μ_{cp}	1	27.89
μ_{cp}	4	26.98
μ_{cp}	8	26.25
μ_{cp}	12	26.37
CPI_QoQ_Ann	1	9.16
CPI_QoQ_Ann	4	9.19
CPI_QoQ_Ann	8	9.14
CPI_QoQ_Ann	12	8.95
Ind_Value_Added_YoY	1	10.85
Ind_Value_Added_YoY	4	12.11
Ind_Value_Added_YoY	8	12.04
Ind_Value_Added_YoY	12	12.03
epu_qavg	1	11.51
epu_qavg	4	15.56
epu_qavg	8	18.69
epu_qavg	12	19.81
gpr_qavg	1	18.74
gpr_qavg	4	20.18
gpr_qavg	8	19.18
gpr_qavg	12	18.49

FEVD shows the percentage contribution of the diagnostic expectations shock to forecast error variance of each variable. Based on median of 500 posterior draws.

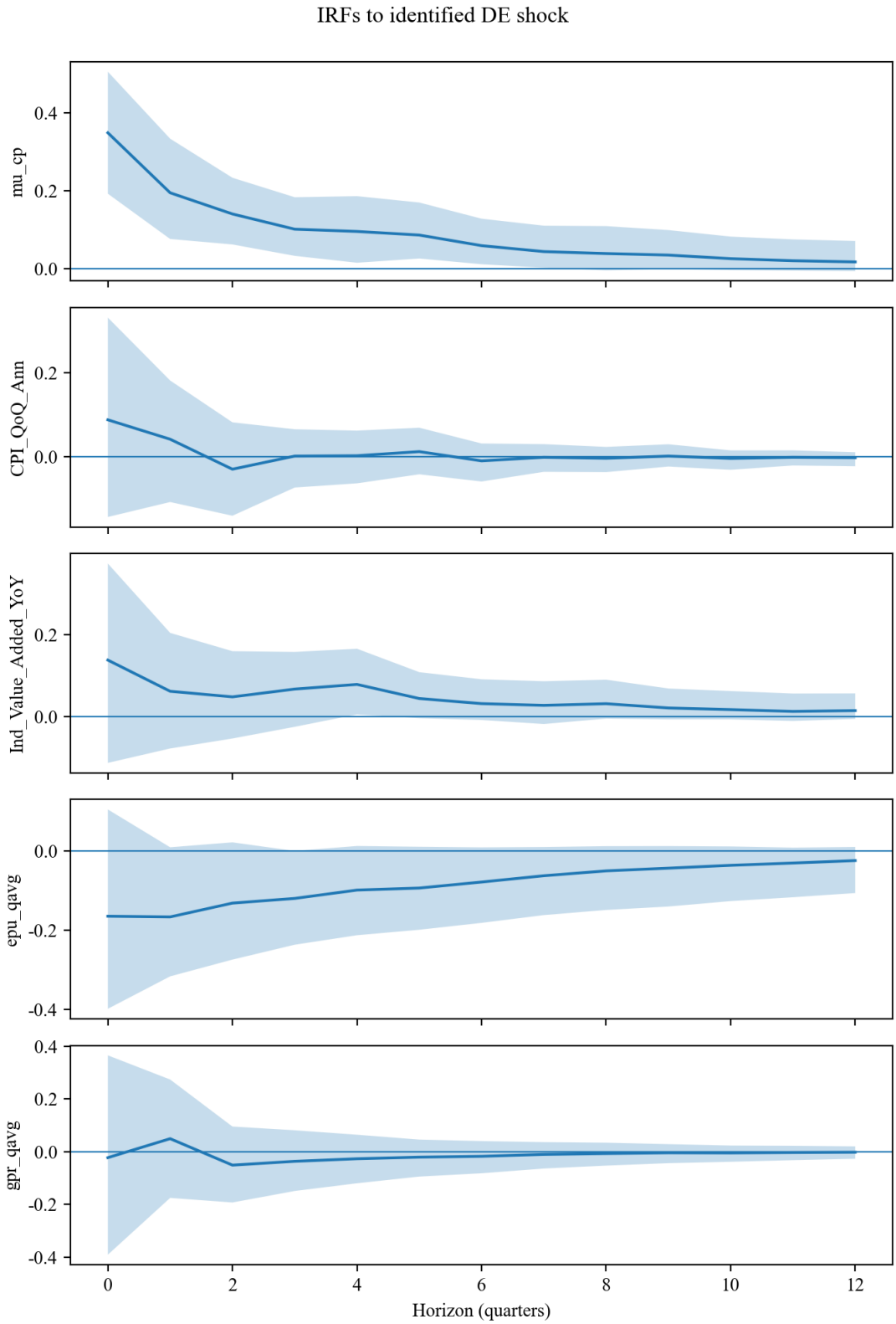

Figure 4: Impulse Responses to a Diagnostic News Shock

Note: Median response (solid line) with 68% credible band (shaded). Responses are in standardized units and correspond to a one-standard-deviation shock.

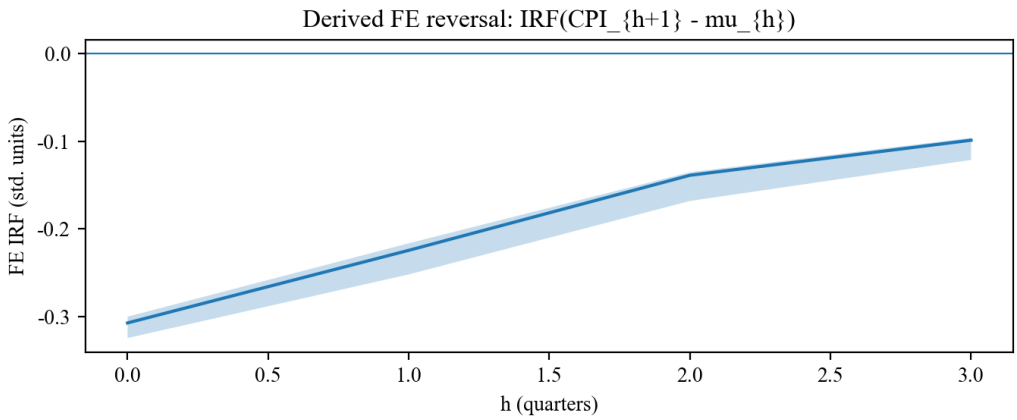

Figure 5: Implied Forecast Error Path Following the Shock
Note: The path of $FE_h = IRF_{\pi^{QoQ,ann}}(h+1) - IRF_{\mu}(h)$. The sustained negative values illustrate the overreaction-correction pattern.

7 Discussion and Implications

Our empirical analysis yields a consistent descriptive pattern: the mean diagnostic coefficient is negative, and the set-identified diagnostic shock reproduces the overreaction-correction dynamic. The median output response is positive but imprecise, so the expansionary-news interpretation is suggestive rather than definitive.

7.1 Interpretation and Alternative Explanations

The core reduced-form finding—a negative β_t—is robust. While we have interpreted it through the lens of diagnostic expectations, it is prudent to consider alternatives. Could it reflect a misspecified learning rule or a particular form of adaptive expectations? A simple adaptive model would typically generate positive serial correlation in forecast errors, not the negative correlation we find. The pattern is more consistent with overreaction than with sluggish adjustment.

Our structural identification strategy is designed to isolate a shock with the specific dynamic signature of diagnostic overreaction to news. Because we do not impose sign restrictions on output, the positive median output response should be interpreted ex post as

suggestive evidence consistent with an 'expansionary news' interpretation, rather than as an identifying restriction. The median responses show declines in uncertainty, though credible bands include zero, so we interpret this channel as suggestive rather than definitive. While no identification scheme is perfect, the constellation of responses provides a coherent picture that aligns well with the theoretical mechanism.

7.2 Policy Implications for Monetary Authorities

The presence of diagnostic biases has important implications for monetary policy, especially communication and credibility management.

First, central bank communication must account for the public's tendency to overreact to salient information. During high-uncertainty episodes or sharp food price swings, clear and calibrated communication from the PBoC could help temper extrapolative expectations. This involves not just stating the inflation target, but explaining why certain price movements are likely transitory, thereby reducing their "representativeness."

Second, policymakers should be cautious in reacting to short-term surges in survey-based inflation expectations. Our variance decomposition indicates that such swings often reflect diagnostic noise with limited pass-through to actual inflation. A premature monetary tightening in response could unnecessarily depress economic activity. Instead, monitoring the underlying drivers of expectations and distinguishing between de-anchoring and temporary overreaction is crucial.

Third, the state-dependent nature of the bias suggests that communication strategies might need to be more active during crises. When β_t is highly negative (e.g., during COVID-19), the public is more prone to overinterpret news. In these windows, more frequent, simple, and direct communication may be necessary to anchor expectations effectively.

7.3 Limitations and Future Research

Our study has limitations that point to fruitful research directions. First, the sample comprises 43 quarterly observations (2013Q4-2025Q2). While this span covers significant macroeconomic events, the degrees of freedom are limited relative to the complexity of a TVP-SSM with time-varying β_t. Given the short sample, weakly informative priors provide needed regularization to prevent overfitting and keep inference stable. Appendix B shows that the main conclusions are robust to looser and tighter prior scales. We acknowledge that the short sample period limits the degrees of freedom, potentially influencing the precision of TVP estimates. However, the BVAR results, which rely on coarser structural features rather than parameter-by-parameter time variation, corroborate the time-varying evidence. Moreover, our Monte Carlo convergence diagnostics (Table 8) show stable posterior inference, suggesting the Bayesian prior structure provides adequate regularization despite the limited sample.

Second, the linear state equation does not find a significant link from EPU/GPR to β_t; the scatter plots are descriptive only. Our data contain only a handful of extreme uncertainty episodes, and we therefore do not implement formal nonlinear or threshold tests. Future work could employ threshold or smooth transition models to formally test this hypothesis, or exploit cross-sectional variation from micro-level household survey data to increase statistical power.

Third, the Carlson-Parkin quantification assumes a fixed indifference threshold $\delta = 0.5$ percentage points. Appendix Table 14 reports a focused sensitivity check with $\delta \in \{0.3, 0.5, 0.7\}$. The summary statistics and OLS/TVP estimates remain negative across these values, so the direction of the diagnostic signature is broadly consistent, but the diagnostic signal weakens as δ rises (e.g., the FE–FR correlation and the share of quarters with negative β_t shrink). We therefore emphasize the qualitative signature rather than precise magnitudes, which are sensitive to the calibration.

Fourth, while we control for food CPI in our measurement equation, we have not explic-

itly isolated supply-driven shocks such as those originating from China's well-known "pork cycle." A pure pork supply shock could mechanically generate forecast errors if households extrapolate temporary price spikes. Future research could specifically disentangle food supply shocks from demand-driven diagnostic expectations using granular micro-data on sectoral price movements and household-level survey responses, potentially employing narrative sign restrictions that explicitly rule out supply-shock interpretations.

Finally, the Chinese context, with its specific media environment and policy communication style, offers a rich setting to explore how institutional factors moderate behavioral biases. Integrating diagnostic expectations into a full-scale New Keynesian model of the Chinese economy would allow for counterfactual policy simulations and welfare analysis. Such extensions remain important avenues for future research.

8 Conclusion

This paper provides evidence consistent with diagnostic expectations as a key feature of household inflation forecast formation in China. Using quarterly survey data and a combination of Bayesian time-varying parameter and structural VAR methods, we document a systematic pattern of overreaction: upward revisions in expectations are followed by negative forecast errors. The intensity of this bias varies over time, with more negative mean estimates after 2018 and in the late sample, though we interpret the path descriptively given wide credible intervals.

By carefully aligning the horizon of expectations and realizations and by identifying a set-identified "diagnostic news shock," we move beyond reduced-form correlations to characterize the dynamic responses implied by the restrictions. The median IRFs imply a large, persistent increase in expected inflation but only a muted response in realized inflation, leading to a prolonged period of negative forecast errors. Median responses are also consistent with rising output and falling uncertainty, though the latter is imprecisely estimated.

Our findings contribute to the growing literature integrating behavioral elements into macroeconomics and offer specific insights for policy in the world's second-largest economy. They underscore that effective monetary policy communication must be designed with an understanding of how households actually process information—sometimes over-weighting the salient and surprising. For China, as its monetary policy framework continues to modernize, recognizing and managing these diagnostic biases will be an important component of maintaining macroeconomic stability.

A Mechanical Coupling Bias: A Sanity Check

Let $\mu_t = \mu_t^* + e_t$ with iid measurement error. Under the rational-expectations null where $\mathrm{Cov}(\pi_{t+1} - \mu_t^*, \mu_t^* - \mu_{t-1}^*) = 0$, the OLS slope in the FE–FR regression is

$$\beta_{\text{mech}} = -\frac{\mathrm{Var}(e_t)}{\mathrm{Var}(FR_t)}, \qquad \mathrm{Var}(FR_t) = \mathrm{Var}(\Delta\mu_t^*) + 2\mathrm{Var}(e_t),$$

so $\beta_{\text{mech}} \in [-0.5, 0)$. This bound relies on iid measurement-error assumptions; serial correlation in e_t could loosen it, so we treat it as a conservative benchmark. Table 12 reports the sample variance of FR_t and the implied mechanical bound alongside the bivariate OLS estimate. The upper bound is much smaller in magnitude than the OLS and TVP estimates, indicating that mechanical coupling alone cannot plausibly explain the observed coefficients.

B Prior Sensitivity

We assess sensitivity of the TVP-SSM to alternative prior scales for the variance parameters and the initial state. The baseline priors use $R, Q \sim \mathcal{IG}(2, 1)$ and $\beta_0 \sim \mathcal{N}(0, 1)$. We compare this to a looser setting with larger scale (Inv-Gamma(2,4), β_0 variance 4) and a tighter setting with smaller scale (Inv-Gamma(2,0.25), β_0 variance 0.25). Table 13 summarizes the resulting

Table 12: Mechanical Coupling Bias: Upper-Bound Check

Statistic	Value
N (FE/FR sample)	43
Var(FR)	0.257
OLS beta (FE on FR, HAC4)	-1.200
Max mechanical beta	-0.500
Max Var(e)	0.129
Var(e) needed for β_{OLS}	0.309
Needed/Max Var(e)	2.40

Under $\mu_t = \mu_t^* + e_t$ with iid e_t, $\beta_{mech} = -\text{Var}(e_t)/\text{Var}(FR_t)$.
Since $\text{Var}(FR_t) = \text{Var}(\Delta\mu_t^*) + 2\text{Var}(e_t)$, $\beta_{mech} \in [-0.5, 0)$.
OLS beta is the bivariate slope (same point estimate as HAC(4)).

β_t path statistics and a simple fit metric.

Table 13: Prior sensitivity for the TVP-SSM

Setting	Beta mean	Beta range	Share $\beta < 0$	LogLik
Baseline	-1.439	[-2.64, -0.09]	100.0%	-107.073
Loose priors	-1.514	[-3.78, 1.08]	83.7%	-104.846
Tight priors	-1.261	[-1.59, -0.14]	100.0%	-108.186

Baseline priors: $R, Q \sim InvGamma(2, 1)$, $\beta_0 \sim \mathcal{N}(0, 1)$.
Loose priors: $R, Q \sim InvGamma(2, 4)$, $\beta_0 \sim \mathcal{N}(0, 4)$.
Tight priors: $R, Q \sim InvGamma(2, 0.25)$, $\beta_0 \sim \mathcal{N}(0, 0.25)$.
LogLik is total-sample log-likelihood at posterior means and the β_t mean path.
FE/FR are in pp units; controls are standardized as in the TVP-SSM.

C Delta Sensitivity

Appendix Table 14 reports how the CP quantification results vary with the indifference band δ.

Table 14: Delta Sensitivity for Carlson-Parkin Quantification

δ	N	Mean μ	SD μ	Mean FE	SD FE	Mean FR	SD FR	Corr(FE,FR)	OLS (SE)	TVP mean β	Share $\beta < 0$
0.3	43	0.286	0.388	1.243	3.183	-0.033	0.446	-0.324	-2.433 (0.674)	-2.486	100.0%
0.5	43	0.400	0.536	1.130	3.171	-0.051	0.507	-0.192	-1.325 (0.729)	-1.439	100.0%
0.7	43	0.513	0.699	1.017	3.169	-0.068	0.602	-0.096	-0.617 (0.722)	-0.814	72.1%

Summary stats use the OLS sample (non-missing FE, FR, and controls).
$FE_t = CPI_{t+1}^{QoQ,ann} - \mu_t$ and $FR_t = \mu_t - \mu_{t-1}$.
OLS includes Food CPI, M2, and PPI with HAC(4) standard errors.
TVP metrics use the posterior-mean β_t path from the TVP-SSM.

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